

## Customer Journey Map

### 3. PROJECT DESIGN PHASE

|               |                                       |
|---------------|---------------------------------------|
| Date          | 4 July 2026                           |
| Team ID       | XXXXXX                                |
| Project Name  | A Comprehensive Measure of Well Being |
| Maximum Marks | 4 Marks                               |

## Proposed Solution Document

This document outlines the design and functional characteristics of the proposed Human Development Index (HDI) Predictor system.

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### Overview of the Solution

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The HDI Predictor is a web-based, machine-learning-powered platform that enables policy analysts, academics, and the public to estimate a country's development index based on key health, education, and income metrics. It offers a fast, interactive simulation sandbox that bypasses manual spreadsheet consolidation.

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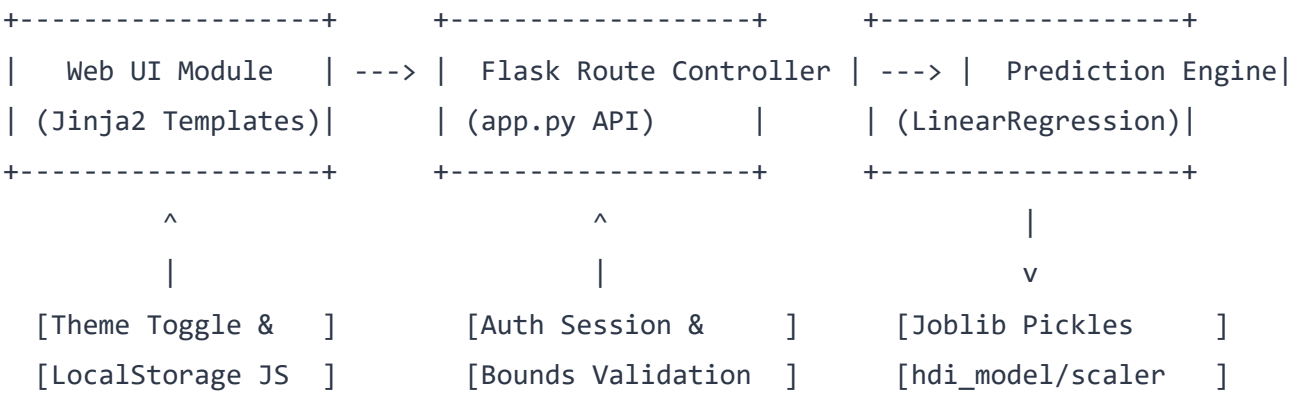
### Design Theme & Core UX Aesthetics

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- Glassmorphism:** Visual elements (cards, headers, sidebars) use translucent backdrops (`backdrop-filter: blur(12px)`) and thin borders (`1px solid rgba(226, 232, 240, 0.8)`) to create depth and modern appeal.

- 2. **Light / Dark Mode:** Custom CSS variables define colors for both themes. Toggle state is preserved in the browser's local cache.
- 3. **Responsive Side Layout:** The desktop dashboard features a sticky sidebar. On mobile viewports, the sidebar collapses into a slide-out drawer, optimizing screen space.
- 4. **Data Visualizations:** Visual correlation heatmaps, pair plots, and strip plots are rendered dynamically under the EDA section, keeping statistical insights close to the calculator.

## ⚙️ Core Application Modules



1. **Data Pipeline:** Ingests UNDP time-series CSV data, cleans missing entries, and fits a StandardScaler.
2. **Predictive Engine:** Linear Regression model mapping feature standard deviations to estimated HDI values.
3. **Web Server:** Flask app serving authentication checks, metrics endpoints, prediction calculators, and API routers.
4. **Interactive Dashboard:** Form controls with synchronized range sliders, dynamic country auto-fill, and client-side history lists with CSV exports.